Backup & Restore Procedure

Ref: 4.4
Product: iConnect Plus
Organisation: Cache Digitech
Classification: Internal — Confidential
Version: 2.0
Date: 2026-11-03
Deployment model: On-premises (Cache Digitech premises) · VPN for remote access
Audience: IT operations, platform engineering, DR coordinator

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1. Purpose

Define **on-premises backup** and **restore** procedures for iConnect Plus (PostgreSQL, MinIO, configuration secrets metadata). **Primary backup target is not AWS RDS/S3** in this deployment model.

2. Scope

In scope	Out of scope
`erp-postgres` logical dumps (`pg_dump` / `pg_dumpall` strategy)	SaaS vendor backup SLAs
MinIO bucket/volume copy	End-user workstation backup
Docker volume snapshots (host-level)	GitHub source (separate process)
`.env` backup (encrypted store)	Optional OpenSearch indices — **Template / Pending sign-off**

3. Backup objects

Object	Method	Frequency (target)	Retention (target)	Storage location
PostgreSQL OLTP	`pg_dump -Fc` or equivalent	Daily full; **Template / Pending sign-off** WAL	**Template / Pending sign-off**	On-prem backup server / NAS
MinIO data	`mc mirror` or volume snapshot	Daily	Match DB policy	On-prem
Redis	Optional RDB — non-authoritative	Weekly or none	Short	On-prem
RabbitMQ	Definitions export optional	Weekly	Short	On-prem
TLS certs + nginx config	File copy	On change	1 year	Secure IT store
Application `.env`	Encrypted archive	On change	2 versions min	IT vault

Actual schedule owner: Cache Digitech IT — **Template / Pending sign-off**.

4. RTO / RPO design targets

Metric	Design target	Validated
RPO (max acceptable data loss)	≤ **24 hours** for OLTP (daily backup baseline)	**Template / Pending sign-off**
RTO (time to restore service)	≤ **8 hours** for full stack on replacement host	**Template / Pending sign-off**
MinIO RPO	≤ **24 hours** (aligned with DB)	**Template / Pending sign-off**

Tighter RPO requires WAL archiving or more frequent dumps — document if implemented.

5. Backup procedure (PostgreSQL)

Prerequisites: VPN or LAN admin access; Docker host SSH; credentials in break-glass vault.

```
# Example — adjust container name and paths per environment
docker exec erp-postgres pg_dump -U "${POSTGRES_USER}" -Fc "${POSTGRES_DB}" \
  > "/backup/erp/$(date +%Y%m%d)/erp_$(date +%Y%m%d_%H%M).dump"
```

Step	Action
1	Verify disk space on backup target
2	Run dump during low-traffic window or use snapshot consistency method
3	Verify dump file size / `pg_restore --list` smoke check
4	Copy to secondary on-prem location (DR shelf / second NAS)
5	Log job success in ops ticket / backup log

6. Backup procedure (MinIO)

```
# Example using MinIO client against internal endpoint
mc alias set erp-local https://127.0.0.1:9000 "$MINIO_ROOT_USER" "$MINIO_ROOT_PASSWORD"
--insecure
mc mirror erp-local/cache-erp-bucket "/backup/erp/minio/$(date +%Y%m%d)/"
```

Bucket name defaults per `S3_BUCKET` / inventory (1.5).

7. Restore procedure (PostgreSQL)

Step	Action	Owner
1	Declare incident / maintenance; stop API and workers	Ops
2	`docker compose stop erp-api erp-celery-worker erp-web`	Ops
3	Restore to **new** volume or rename old volume	Ops
4	`pg_restore` or recreate DB from dump	Ops
5	Run Alembic `upgrade head` if dump is behind schema	Platform
6	Start stack; smoke test login + critical transaction	Platform + business
7	Record in restore test report (below)	Ops

Data loss window: Between last good backup and failure time — communicate to stakeholders.

8. Restore procedure (MinIO)

Restore mirror to `erp_minio_data` volume or re-point bucket after volume restore. Verify document download and virus scan path (ClamAV).

9. Restore test template

Field	Value
Test ID	**Template / Pending sign-off**
Date	
Environment	UAT / DR host (not prod unless approved)
Backup used	File name + timestamp
Performed by	
PostgreSQL restore	Pass / Fail — notes
MinIO restore	Pass / Fail — notes
App smoke test	Pass / Fail — login, one read, one write
RTO measured	hours
Issues	
Sign-off (IT)	
Sign-off (Platform)	

Last completed restore test: Template / Pending sign-off

10. Security of backups

Control	Requirement
Encryption at rest	**Template / Pending sign-off**
Access	IT break-glass only
Offsite	Second premises location preferred over public cloud
Classification	Same as production Confidential

11. Related documents

Ref	Document
5.5	BCP / DR Plan & DR Test Report
5.6	Operations Runbook
1.5	Asset & Component Inventory
4.3	Data Retention & Deletion Schedule

12. Change log

Date	Change	By
2026-11-03	v2.0 on-prem backup baseline	Platform